



Advanced Exploration Systems Life Support Systems

Acquisition of Temperature and Humidity Equipment System

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Sorbent-Based Atmospheric Revitalization (SBAR)

George C. Marshall Space Flight Center
Engineering Directorate, Space Systems Department

Objective/Requirements

The Environmental Control and Life Support Systems (ECLSS) team at Marshall Space Flight Center was provided funding through for the advanced testing and development of the Sorbent Based Atmospheric Revitalization (SBAR) system. The SBAR system removes humidity and carbon dioxide (CO₂) to enable habitation in low pressure environments.

To evaluate predicted SBAR performance in cabin scenarios, the ground test system must utilize a temperature and humidity system to simulate inlet air conditions in the full range of cabin environments, as well as post-crew dry out scenarios. As part of this effort, a highly capable, reliable temperature and humidity conditioning system must be procured that can meet the wide envelopes of temperatures and humidity at various flow rates.

Characteristics, Scope, and Specs

The contractor shall provide a temperature and humidity conditioning system that meets the following specs:

- Exterior dimensions no larger than 54" wide x 50" deep x 76" high
- Warranty with customer support. Manuals, schematics, and drawings on flash drive.
- Safety features to prevent over temperature and hi/low pressure events
- Fuses, circuit breakers and thermal overloads to protect circuits and components
- System monitoring and preventative maintenance settings
- Contains non-explosive, non-ozone depleting refrigerants in compliance with EPA SNAP regulations
- Electrical Service: 460 volt, 3 phase, 60 hertz power supply needed
- Control Accuracy: $\pm 1^{\circ}\text{C}$ ($\pm 2^{\circ}\text{F}$) dry bulb air temperature control tolerance
- Humidity Stability: $\pm 5\%$ RH controlled at the outlet of the conditioner for all conditions above $+45^{\circ}\text{F}$ dewpoint, and $\pm 10\%$ RH for all conditions below a $+45^{\circ}\text{F}$ dewpoint.
- Main power disconnect switch and E-stop (control panel mounted)
- Touch screen control mounted no higher than 48 inches
- User Definable Graphical Trending of Program & Process Data Real Time
- Data logging and data export capability
- Solid state humidity sensor
- Inline Thermal Mass Flowmeter
- Evaporator pressure regulator
- Auto-switchover of supply air source based on setpoints (Missile grade or fresh air)

- Product temperature control used to control to a remote temperature sensor located at the point of use, while controlling & limiting the unit temperature to prevent overshoot and/or oscillation during varying flow rates or during flow interruptions
- Capable of providing conditioned air at 6 – 35 scfm
- Capable of providing conditioned air at the following specifications:

Dewpoint	Temperature	Cooling Reg	Flow Rate	RH
[F]	[F]	[F]	[SCFM]	[%]
73.0	78.0	13.6	7.2	84.7
62.2	67.2	1.9	6.3	84.0
43.9	66.4	0.0	7.9	44.2
80.5	85.5	13.9	6.7	85.1
50.6	73.4	0.0	9.3	44.8
87.4	92.4	14.2	7.7	85.5
76.2	81.2	1.7	7.0	84.8
57.2	80.6	0.0	10.4	44.8
66.2	72.0	0.0	6.3	82.1
Dewpoint	Temperature	Cooling Reg	Flow Rate	RH
[F]	[F]	[F]	[SCFM]	[%]
87.4	92.4	14.2	6.3	85.5
80.5	85.5	7.2	6.3	85.1
71.8	78.3	0.0	6.3	80.6
60.1	78.3	0.0	6.3	53.7
40.7	78.3	0.0	6.3	26.1
-20.2	78.3	0.0	6.3	1.7

Dewpoint	Temperature	Flow Rate	RH
[F]	[F]	[SCFM]	[%]
18.3	65.0	14.9	16.4
36.2	65.0	14.9	34.3
18.3	80.0	14.5	9.9
36.2	80.0	14.5	20.7
42.0	70.0	13.3	36.3
42.0	80.0	13.0	26.0
38.0	70.0	31.4	31.0
38.0	80.0	30.8	22.2
48.0	70.0	27.9	45.5
55.0	76.0	27.6	48.2
48.0	70.0	27.9	45.5
55.0	76.0	27.6	48.2
51.7	70.0	9.2	52.2
51.7	80.0	9.0	37.4
47.5	70.0	21.8	44.7
47.5	80.0	21.4	32.0
57.9	70.0	19.3	65.6
65.3	70.3	19.3	84.2
57.9	76.0	19.1	53.6
65.3	76.0	19.1	69.4

Performance Requirements and Deliverables:

Marshall Space Flight Center, Huntsville, AL

Period of Performance:

The unit must be delivered to Marshall Space Flight Center, Huntsville, AL no later than 7 months after award.